

Probabilistic Analysis and Quick Sort Notes

Introduction

This technical note summarizes the tools and logic used in probabilistic analysis.

Type of probabilistic analysis

There are four main types of probabilistic analysis.

- Random if
 - With Recursion
 - Without recursion
- Random for
 - single random loop
 - Random loop inside an outer loop
- Random-sized recursive call (i.e. Number line problem)
- Loop that randomly terminates

1. Random if

Method: *Find the probabilities of both conditions and their corresponding $T(n)$, then use the formula.*

1.1 Random if with recursion

```

1. k = Random(n)
2. if Boolean condition on k then
3.     Make recursive call
4. else
5.     Make recursive call
6. end if

```

For example

```

function Function1(A[], n) {
    k = random(n)
    if (k mod 4 = 1) {
        x = x + Function1(A[], n/2)
    } else {
        x = x + Function1(A[], n/3)
    }
}

```

We can use the following formula:

$$ET(n) = \Pr(A) * E(T(n) | A) + \Pr(\text{Not } A) * E(T(n) | \text{Not } A)$$

OR

$$ET(n) = \Pr(A) * E(T(n) \mid A) + \Pr(\text{Not } A) * E(T(n) \mid \neg A)$$

From the formula, we can easily see that

$$ET(n) = \frac{1}{4} * T\left(\frac{n}{2}\right) + \left(1 - \frac{1}{4}\right) * T\left(\frac{n}{3}\right)$$

1.2 Random if without recursion

```

1. k = Random(n)
2. if Boolean condition on k then
3.     Foo()
4. else
5.     Bar()
6. end if

```

For example

```

function Function2(A[], n) {
    k = random(n)
    int x = 0;
    if (k mod 4 = 1) {
        for (i = 1 to n) {
            x = x + A[i];
        }
    } else {
        int i = 1;
        while (i <= n) {
            x = x + A[i];
            i = 2 * i;
        }
    }
}

```

We can use the following formula (Same as in Section 1.1):

$$ET(n) = \Pr(A) * E(T(n) \mid A) + \Pr(\text{Not } A) * E(T(n) \mid \neg A)$$

OR

$$ET(n) = \Pr(\text{Foo}) * E(T(n) \mid \text{Foo}) + \Pr(\text{Bar}) * E(T(n) \mid \text{Bar})$$

Since

$$\Pr(A) = \frac{1}{4}, E(T(n) \mid \text{Foo}) = \Theta(n)$$

and

$$\Pr(\neg A) = \frac{3}{4}, E(T(n) \mid Bar) = \Theta(\log n)$$

From the formula, we can easily get that

$$ET(n) = \frac{1}{4} * \Theta(n) + \frac{3}{4} * \Theta(\log n)$$

2. Random for

Method: *First analyze the running time for a fixed random value of k , and then incorporate the result into the outer loop if there is one.*

2.1 single random loop

1. $k = \text{Random}(n)$
2. Loop that runs $f(k)$ times

For example

```
function Function3(A[], n) {
    x = 0;
    k = Random(n);
    for (i = 1 to k){
        for(j = 1 to i * i) {
            x = x + A[ij mod n];
        }
    }

    return x;
}
```

We can use the formula:

$$ET(n) = \sum_{q=1}^n \Pr(k = q)t(k = q)$$

We should first find $t(k=q)$, which is the running time of the loop controlled by k .

$$t(k = q) = \left(\sum_{i=1}^q i^2 C \right)$$

Since k is uniformly random,

$$\Pr(k = q) = \frac{1}{n}$$

Therefore,

$$ET(n) = \sum_{q=1}^n \frac{1}{n} \sum_{i=1}^q i^2 C$$

2.2 Random loop inside an outer loop

1. loop
2. k = Random(i)
3. Loop that runs f(i) times
4. end loop

For example

```
function Function4(A[], n) {
    x = 0;
    for (i = 1 to n) {
        k = Random(i);
        for (j = 1 to k * k){
            x = x + A[(i*j mod n)]
        }
    }

    return x;
}
```

For each iteration of the outer loop, we first analyze the time complexity of the inner loop. (Same as what we did in Section 2.1)

We can use the formula:

$$ET_i(n) = \sum_{q=1}^i \Pr(k = q) t(k = q)$$

Since k is uniformly random from 1 to i ,

$$\Pr(k = q) = \frac{1}{i}$$

and because the inner loop runs k^2 times,

$$t(k = q) = q^2 C$$

Therefore,

$$\begin{aligned} ET_i(n) &= \sum_{q=1}^i \frac{1}{i} \Theta(q^2) \\ &= \frac{1}{i} \Theta(i^3) = \Theta(i^2) \end{aligned}$$

Then, we can determine the overall time complexity.

$$ET(n) = \sum_{i=1}^n ET_i(n)$$

$$= \sum_{i=1}^n \Theta(i^2) = \Theta(n^3)$$

3. Random-sized recursive call (i.e. Number Line problem)

Method: *First identify the best and worst cases. Then use a number line to divide the possible values of k into ranges, and use the appropriate worst-case bound in each range to determine the overall time complexity.*

1. Base case and non-recursive work
2. Do ()
3. $k = \text{Random}(n)$
4. Make recursive call of size $f(k)$.
4. return

For example

```
function Function5(A[], n) {
    if (n <= 22) {
        return 1;
    }
    x = 0;
    for (i = 1 to n/2) {
        for (j = 1 to n/4) {
            x = x + A[(ij mod n)];
        }
    }
    k = Random(n/5 - 1);
    x = x + Func5(A[], 5k);
    x = x + Func5(A[], n - 5k);

    return x;
}
```

We use the following steps:

1. Substitute the minimum and maximum possible values of k and compare the resulting recursive-call sizes to identify the best and worst cases.
2. If the minimum and maximum values of k produce the same case, also test the midpoint

$$k = \frac{k_{\min} + k_{\max}}{2}$$

to determine whether it gives the best case.

Range 1

Range 2

Range 3

Range 4

For the upper bound, choose the worst recursive-call sizes in each range.

Therefore,

$$ET(n) \leq C_1 n^2 + \frac{1}{4} [ET(5) + ET(n-5)] + \frac{1}{4} [ET(n/4) + ET(3n/4)] \\ + \frac{1}{4} [ET(3n/4) + ET(n/4)] + \frac{1}{4} [ET(n-5) + ET(5)].$$

Combining identical terms gives

$$ET(n) \leq C_1 n^2 + \frac{1}{2} [ET(5) + ET(n-5)] + \frac{1}{2} [ET(n/4) + ET(3n/4)].$$

Since $ET(5) = \Theta(1)$, it can be absorbed into the other asymptotic terms.

For the upper bound, we can also replace $ET(n-5)$ by $ET(n)$ because

$$n-5 < n$$

and therefore

$$ET(n-5) \leq ET(n).$$

Thus,

$$ET(n) \leq C_2 n^2 + \frac{1}{2} ET(n) + \frac{1}{2} ET(n/4) + \frac{1}{2} ET(3n/4)$$

Move $\frac{1}{2} ET(n)$ to the left-hand side:

$$ET(n) - \frac{1}{2} ET(n) \leq C_2 n^2 + \frac{1}{2} ET(n/4) + \frac{1}{2} ET(3n/4)$$

Therefore,

$$\frac{1}{2} ET(n) \leq C_2 n^2 + \frac{1}{2} ET(n/4) + \frac{1}{2} ET(3n/4)$$

Multiplying both sides by 2 gives

$$ET(n) \leq ET(n/4) + ET(3n/4) + Cn^2$$

Therefore, using a recursion tree, we obtain

$$ET(n) = O(n^2).$$

For the Lower Bound, the expected running time cannot be better than the best possible recursive split.

The best split occurs at

$$k = \frac{n}{10},$$

which gives two recursive calls of size $n/2$.

Therefore, the best-case recurrence is

$$T(n) = 2T(n/2) + Cn^2.$$

This recurrence gives

$$T(n) = \Theta(n^2).$$

Since the expected running time cannot be better than the best-case running time,

$$ET(n) = \Omega(n^2).$$

Therefore,

$$ET(n) = \Theta(n^2)$$

4. Loop that randomly terminates

```
1. Do some work
2. repeat
3.     k = Random(n)
4.     Do some work
5. until Boolean condition on k is true
```

For this type of problem, the loop terminates only when a random condition is satisfied.

We let X be the number of times the loop repeats before termination.

For example:

```
function Function6(A[], n) {
    x = 0;

    repeat {
        k = Random(n);

        for (i = 1 to sqrt(n)) {
            x = x + A[i];
        }

    } until (k == 42);

    return x;
}
```

The loop terminates when

$$k = 42.$$

Since k is uniformly random from 1 to n ,

$$\Pr(k = 42) = \frac{1}{n}$$

and

$$\Pr(k \neq 42) = \frac{n-1}{n}.$$

For the expected case, let X be the number of times the loop repeats before termination.

The running time can be written as

$$T(n) = c\sqrt{n} + X \cdot c\sqrt{n}.$$

Therefore,

$$ET(n) = c\sqrt{n} + E(X) \cdot c\sqrt{n}.$$

To find $E(X)$, use

$$E(X) = \sum_{i=0}^{\infty} i \Pr(X = i).$$

Then,

$$E(X) = \sum_{i=1}^{\infty} \Pr(X \geq i).$$

For the loop to repeat at least i times, the terminating condition must fail repeatedly. Thus,

$$\Pr(X \geq i) = \left(\frac{n-1}{n}\right)^i.$$

Therefore,

$$E(X) = \sum_{i=1}^{\infty} \left(\frac{n-1}{n}\right)^i.$$

Using the geometric-series formula,

$$\sum_{i=1}^{\infty} r^i = \frac{r}{1-r},$$

we get

$$E(X) = \frac{\frac{n-1}{n}}{1 - \frac{n-1}{n}}.$$

Since

$$1 - \frac{n-1}{n} = \frac{1}{n},$$

we have

$$E(X) = \frac{n-1}{n} \cdot n = n-1.$$

Thus,

$$ET(n) = c\sqrt{n} + (n-1)c\sqrt{n}.$$

Therefore,

$$ET(n) = cn\sqrt{n}.$$

So,

$$ET(n) = \Theta(n^{3/2})$$

Reference List

1. "CSE2331_Types_of_Probabilistic_Problems" Created by Professor Painter.
2. "Quick Sort Worksheet AU26" Created by Professor Painter.
3. "CSE2331_QuickSort_and_Probabilistic_Analysis_Homework" Created by Professor Painter.